



University of Brighton

The Acute Effect of a DNS-Informed Respiratory-Postural Warm-Up on Running Economy and Ventilatory Responses in Recreational Runners

Introduction and what is the purpose of the study/project?

My name is Alexander Worden and I am a student on the MSc Applied Sport Physiology course at the University of Brighton. I am carrying out this research as part of my MSc dissertation under the supervision of Dr Jake Butterworth.

The purpose of this study is to investigate whether a short Dynamic Neuromuscular Stabilisation (DNS)-informed respiratory-postural exercise sequence influences oxygen use and breathing responses during submaximal treadmill running. The study focuses on recreational runners and examines whether adding a DNS-informed exercise sequence to a standard runner warm-up affects running economy/oxygen cost and breathing responses compared with a time-matched active-neutral control condition.

Invitation:

We would like to invite you to take part in this research study. Before you decide whether to take part, it is important for you to understand why the research is being done and what it would involve for you. Please take time to read the following information carefully. You may discuss the study with others if you wish and ask the researcher if anything is unclear. You are free to take this information sheet away and take time to decide whether you wish to participate.

Why have I been invited to participate?

You have been invited to take part because you are a recreational runner who runs regularly and may meet the eligibility criteria for this study. Approximately 16–20 participants will take part.

You may be eligible to take part if you are:

- an adult aged 18–45 years;
- a recreational runner;
- currently running at least two times per week and have done so for at least six months;
- able to run continuously for 30 minutes;
- able to complete up to approximately 45 minutes of staged treadmill running during the first laboratory visit;
- have a recent or estimated 5 km time of 20–35 minutes;
- free from injury affecting running;
- able to attend three laboratory visits at the University of Brighton Falmer Campus.

You will not be able to take part if you have a condition or circumstance that would make vigorous treadmill exercise, breathing measurements, or fingertip blood sampling unsuitable. This may include relevant cardiovascular, respiratory, neurological, metabolic, or musculoskeletal conditions, current injury affecting running, pregnancy, current smoking or vaping, current use of beta-blockers or respiratory medication, or any other factor that means exercise testing would not be appropriate.

After you have provided written informed consent, you will be asked to complete the ACSM Preparticipation Health Screening Questionnaire through the University laboratory SharePoint system. This is used to confirm suitability for exercise testing. You will not be asked to complete this screening questionnaire before providing consent.

Do I have to take part?

No. Taking part is entirely voluntary. You are free to decline to participate or to withdraw from the study at any time, without giving a reason and without any negative consequences.

What will happen to me if I take part?

If you decide to take part, you will attend three laboratory visits at the University of Brighton Falmer Campus, Meads Sport and Exercise Science Labs, with at least 48 hours between visits.

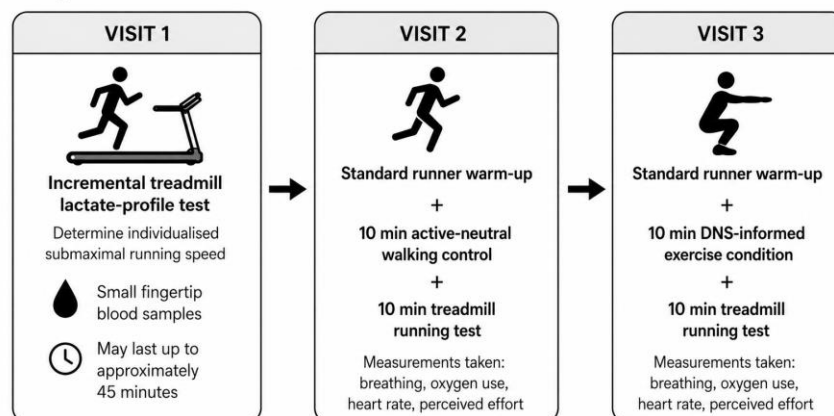
You will complete three visits. The first visit will use an incremental treadmill lactate-profile test to determine the individualised submaximal running speed used in the experimental trials. The following two visits will involve running at that same speed after two different time-matched conditions.

Visit 1 will involve an incremental treadmill lactate-profile test. The treadmill speed will increase in stages, and small fingertip blood samples will be collected during the test to measure blood lactate. This test may last up to approximately 45 minutes, depending on your response to the exercise stages. The information from this visit will be used to determine the individualised submaximal running speed used in Visits 2 and 3. You will also be familiarised with the treadmill, breathing measurement equipment, rating scales, and study procedures.

Visits 2 and 3 will involve treadmill running at the individualised speed determined from Visit 1. Both visits will begin with the same standard runner warm-up. After this, one visit will involve a 10-minute low-intensity walking control condition, and the other will involve a 10-minute DNS-informed respiratory-postural exercise condition consisting of low-load bodyweight exercises. Each condition will then be followed by a 10-minute treadmill running test. During exercise, your breathing, oxygen use, heart rate, and perceived effort will be monitored.

You will be asked to avoid strenuous exercise and alcohol for 24 hours before each visit, avoid caffeine for 3 hours before each visit, consume a similar light meal 2–3 hours before each visit, and wear the same footwear and similar clothing where possible. These factors may influence physiological responses. Exact pre-visit instructions will be provided before testing.

Study overview



Will I be paid for taking part?

No payment or financial compensation will be provided for taking part in this study.

What are the potential disadvantages or risks of taking part?

The study involves staged treadmill running, 10-minute treadmill running tests, breath-by-breath gas exchange measurements using a face mask, heart-rate monitoring, low-load warm-up exercises, low-intensity treadmill walking, and fingertip blood sampling. There is a small risk of fatigue, dizziness, loss of balance, breathlessness, or minor muscle strain during treadmill running or warm-up activities.

Some stages of the incremental treadmill test may feel challenging, but the test is not intended to be a maximal-effort test, and you may stop at any time.

Fingertip blood samples will be collected using sterile, single-use equipment by trained personnel. This may cause brief discomfort, minor bleeding, or slight bruising. All sessions will be supervised, safety procedures will be in place, and you may stop at any time if you feel uncomfortable or unwell.

What are the potential benefits of taking part?

You may request individual feedback on selected testing outcomes, such as your lactate-profile information, individualised running speed, and running economy/oxygen cost measures. The main benefit of taking part is contributing to research on warm-up strategies and breathing-related responses in recreational runners.

The findings may contribute to improved understanding of warm-up strategies and respiratory-postural approaches in running, which may be of benefit to the wider running and sport science community in the future.

Will my taking part in the study/project be kept confidential?

Yes. All data will be treated confidentially. You will be assigned a participant ID code so that your name is not attached to your physiological or perceptual data. The file linking your name to your participant ID will be stored separately from the research data in a secure, password-protected University OneDrive folder. Electronic research data will be stored on University OneDrive only. No research data or personal data will be stored on personal devices, shared laboratory computers, or shared systems. Breath-by-breath gas exchange data will be exported immediately after testing, uploaded to University OneDrive, and deleted from the metabolic system and any associated local or shared laboratory computer.

Preparticipation screening data will be completed through the University laboratory SharePoint system after consent. The researcher will not download or store completed screening questionnaires outside the approved University system.

Only the researcher and supervisor will have access to identifiable or coded research data. Data will be reported only as group-level findings, anonymised values, or coded results. No individual participant will be identifiable in the dissertation.

Confidentiality may be broken if information is disclosed that indicates a risk of harm to yourself or others. In such cases, appropriate University safeguarding procedures will be followed.

Further information about how the University handles research data can be found in the University Research Privacy Notice:

<https://unibrighton.ac.sharepoint.com/:b:/s/public/ERAnx7fZSR9Lv6MRZ1KwpMMBcOSYGp30CAwkTzbbzHLDNQ>



QR code for privacy notice:

What will happen if I don't want to carry on with the study?

You may withdraw from the study at any time without giving a reason. You may also request withdrawal of your data up to **31 August 2026 at 17:00**.

Before this date, your coded data can be identified using your participant ID code and removed from the dataset. After 31 August 2026 at 17:00, your data will have been anonymised and included in group-level analysis for the dissertation, so it will no longer be possible to remove your individual data.

What will happen to the results of the project?

The results will be used for an MSc dissertation submitted to the University of Brighton. The findings will not be published externally or made publicly available without further ethical approval.

All data will be anonymised, and no individual participants will be identifiable. You may request a brief summary of the findings once the project has been completed.

Who is organising and funding the research? (not required for student research where no funding involved)

The research is organised by the University of Brighton as part of an MSc dissertation project. There is no external funding.

Contact details

If you would like more information, please contact:

Alexander Worden
MSc Applied Sport Physiology Student
University of Brighton
Email: A.Worden1@uni.brighton.ac.uk

Supervisor:
Dr Jake Butterworth
University of Brighton
Email: J.Butterworth2@brighton.ac.uk

What if I have a question or concern?

If you have any questions or concerns about the study, you may contact the researcher or supervisor using the details provided above.

If you have any concerns about the ethical conduct of this study, you may contact:

Dr Alex Channon
Chair, Tier 1 Ethics Committee
School of Education, Sport and Health Sciences
University of Brighton

Who has reviewed the study?

This study has been submitted for review by the University of Brighton Research Ethics Committee. Data collection will not begin until a favourable ethical opinion has been granted.